AVRAMOV-TYPE REGULARITY: A PERFECT MAP INTO A TARGET OVER A LOCAL COMPLETE INTERSECTION BASE FORCES BOTH MAPS TO BE COMPLETE INTERSECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18356-18374 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18376-18418, the complete author proof (41 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. This is the scheme-global form of Avramov's theorem that a perfect map with complete-intersection total-to-base map is itself a complete intersection; named by the author via the label.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Equivalence of regular and Koszul-regular sequences in Noetherian rings (More on Algebra, lemma-noetherian-finite-all-equivalent), localization of regularity for ideals (Algebra, lemma-regular-sequence-in-neighbourhood; Divisors, lemma-Noetherian-scheme-regular-ideal), and the divided-powers criterion for perfect maps between regular immersions (Divided Power Algebra, lemma-perfect-map-ci).

Source reference. Original lemma 061L: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18418–18418. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

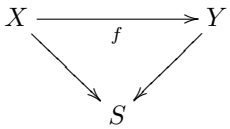
Source reference. Original lemma 063I: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18418–18418. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09PX: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18418–18418. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09CC: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18418–18418. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let*



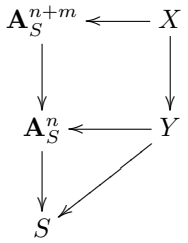
be a commutative diagram of morphisms of schemes. Assume

(1) *S is locally Noetherian,*
(2) *$Y \to S$ is locally of finite type,*
(3) *$f : X \to Y$ is perfect,*
(4) *$X \to S$ is a local complete intersection morphism.*

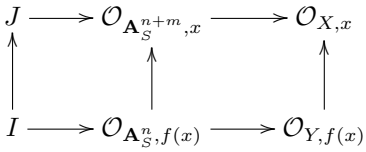
Then $X \to Y$ is a local complete intersection morphism and $Y \to S$ is Koszul at $f(x)$ for all $x \in X$.

Proof. In the course of this proof all schemes will be locally Noetherian and all rings will be Noetherian. We will use without further mention that regular sequences and Koszul regular sequences agree in this setting, see More on Algebra, Lemma 09CC. Moreover, whether an ideal (resp. ideal sheaf) is regular may be checked on local rings (resp. stalks), see Algebra, Lemma 061L (resp. Divisors, Lemma 063I)

The question is local. Hence we may assume S, X, Y are affine. In this situation we may choose a commutative diagram



whose horizontal arrows are closed immersions. Let $x \in X$ be a point and consider the corresponding commutative diagram of local rings



where J and I are the kernels of the horizontal arrows. Since $X \to S$ is a local complete intersection morphism, the ideal J is generated by a regular sequence. Since $X \to Y$ is perfect the ring $\mathcal{O}_{X,x}$ has finite tor dimension over $\mathcal{O}_{Y,f(x)}$. Hence we may apply Divided Power Algebra, Lemma 09PX to conclude that I and J/I are generated by regular sequences. By our initial remarks, this finishes the proof. □